Advanced Machine Learning

Lab 3 --- Module 1

Instructions on how to submit your work on Canvas

To convert your Notebook to an HTML file, follow these steps:

- Open the Notebook that you want to convert to an HTML file.
- Ensure that the notebook is saved with the latest changes.
- In the Notebook interface, navigate to File in the menu bar.
- Click on Download as and select HTML (.html) from the dropdown menu. This will initiate the conversion process.
- The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
- Locate the downloaded HTML file on your computer and verify that the conversion was successful.
- Optionally, open the HTML file in a web browser to view and verify the content.
- Upload your HTML file on Canvas.

Make sure to save your Notebook before attempting to convert it to an HTML file.

Exercise 1 (3 points)

You are a data scientist working on a project involving the optimization of a machine learning model for a critical business application. Your goal is to estimate the parameters of the model to maximize its predictive accuracy while ensuring that it meets certain computational constraints.

The optimization problem involves minimizing a cost function while satisfying a computational constraint. The cost function and constraint are given as follows:

$$\begin{aligned} \text{Minimize:} \quad & J(\boldsymbol{\theta}) = 5\theta_1^2 + 3\theta_2^2 && (1) \\ \text{Subject to:} \quad & g(\boldsymbol{\theta}) = 4\theta_1 + 2\theta_2 - 12 = 0 && (2) \end{aligned}$$

Where:

- $\boldsymbol{\theta} = [\theta_1, \theta_2]$ represents the model parameters of the machine learning model.

- $J(\boldsymbol{\theta})$ is the cost function that depends on the model parameters.
- $g(\boldsymbol{\theta})$ is the computational constraint that must be satisfied during optimization.

1. Formulate the Lagrangian Function:
   - Define the Lagrangian function $\mathrm{L}(\boldsymbol{\theta}, \lambda)$ that combines the cost function and the constraint using Lagrange multipliers λ:

   $$\mathrm{L}(\boldsymbol{\theta}, \lambda) = J(\boldsymbol{\theta}) + \lambda \cdot g(\boldsymbol{\theta})$$

2. Find the Stationary Points:
   - Calculate the gradient of $\mathrm{L}(\boldsymbol{\theta}, \lambda)$ with respect to $\boldsymbol{\theta}$ and set it to zero to find the stationary points:

   $$\nabla_{\boldsymbol{\theta}} \mathrm{L}(\boldsymbol{\theta}, \lambda) = \mathbf{0}$$

   - Solve the system of equations to find the optimal model parameters $\boldsymbol{\theta}$ and Lagrange multiplier λ.
3. Interpret Results:
   - Interpret the values of $\boldsymbol{\theta}$ and λ in the context of model parameter optimization for your machine learning model. What do they represent?

Exercise 2 (2 points)

Create visualizations, such as contour plots, to illustrate the optimal value $\boldsymbol{\theta}$.